

Filter Wizard

filter2

Created on 02/23/2026

**ANALOG
DEVICES**

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Filter Wizard Design Report

Filter Requirements for Band-Pass, 8th order Butterworth

Specifications: Optimize: Noise; +Vs: 3.3; -Vs: 0

Gain: 40 dB

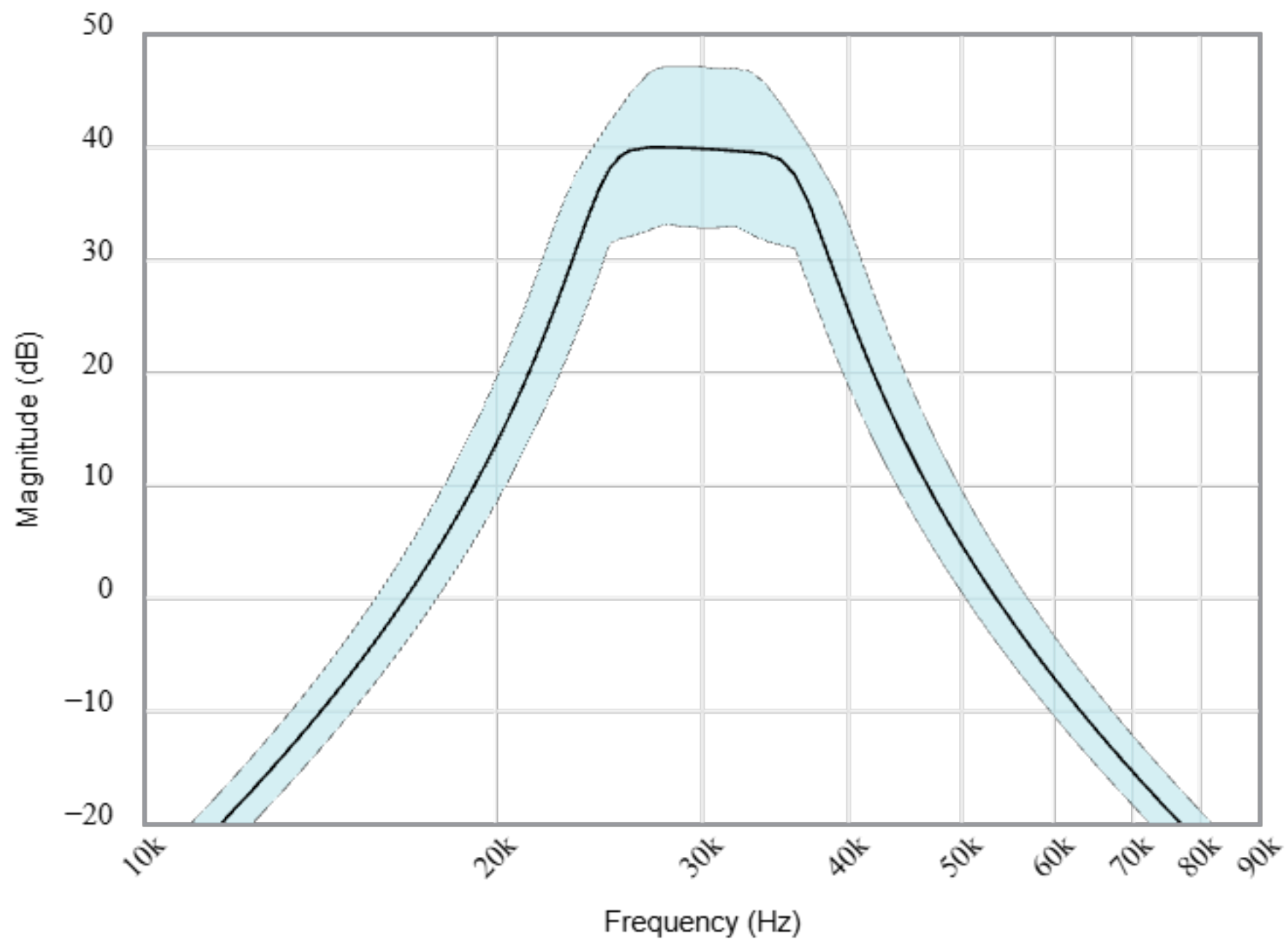
Passband: -1dB at 10kHz

Stopband: -40dB at 40kHz

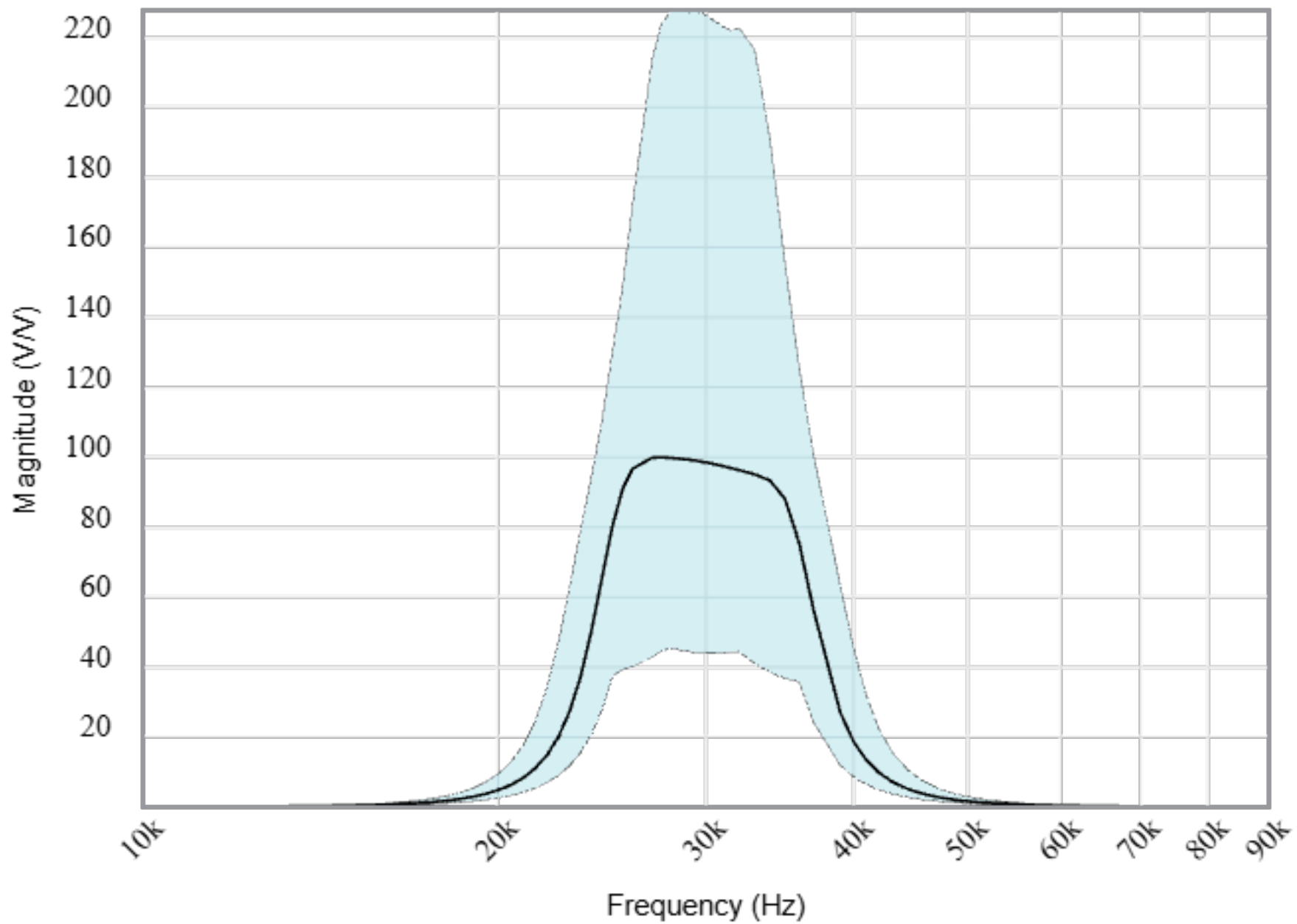
Component Tolerances: Capacitor = 5%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

BOM: refer to BOM.csv file

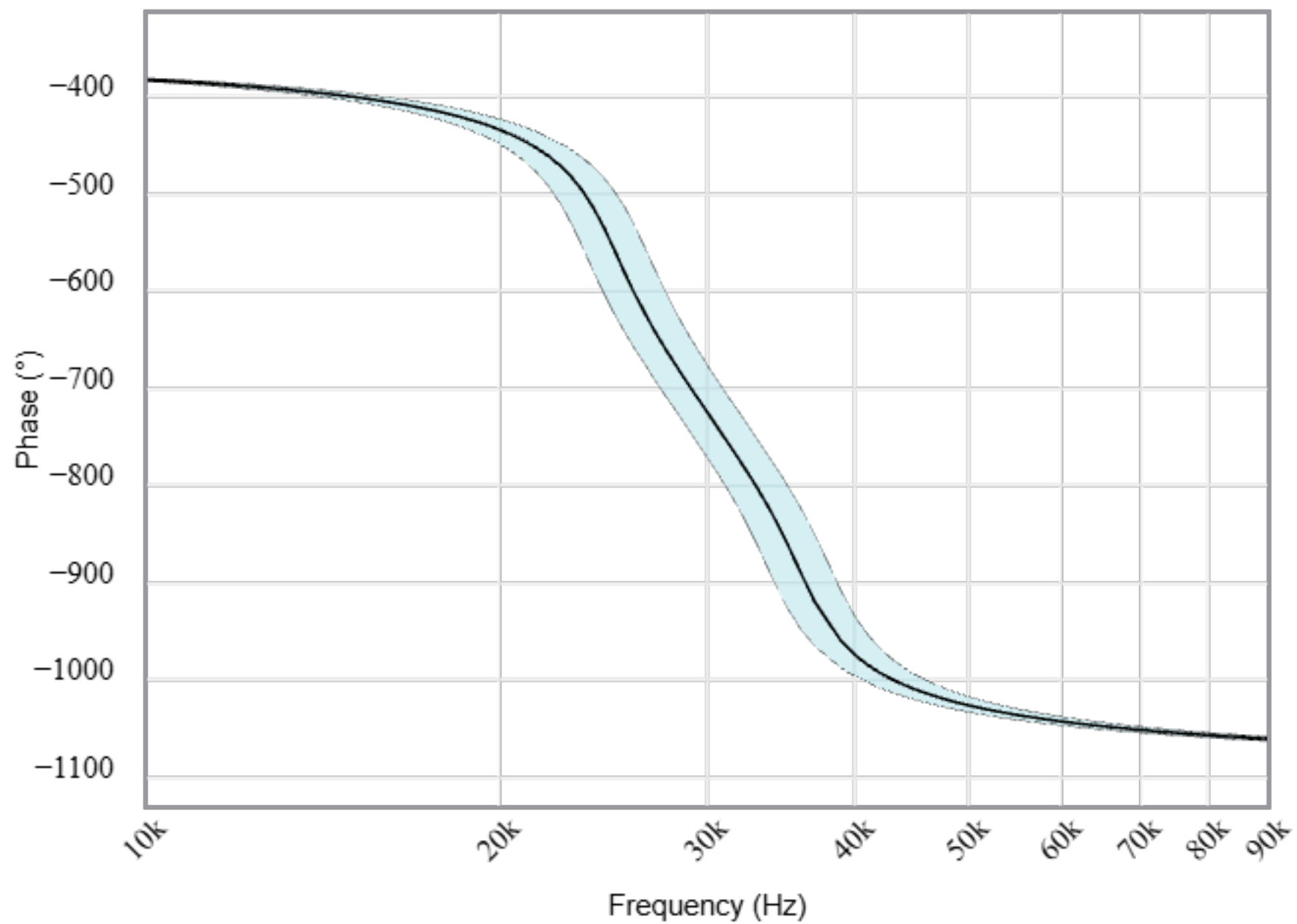
Magnitude(dB)



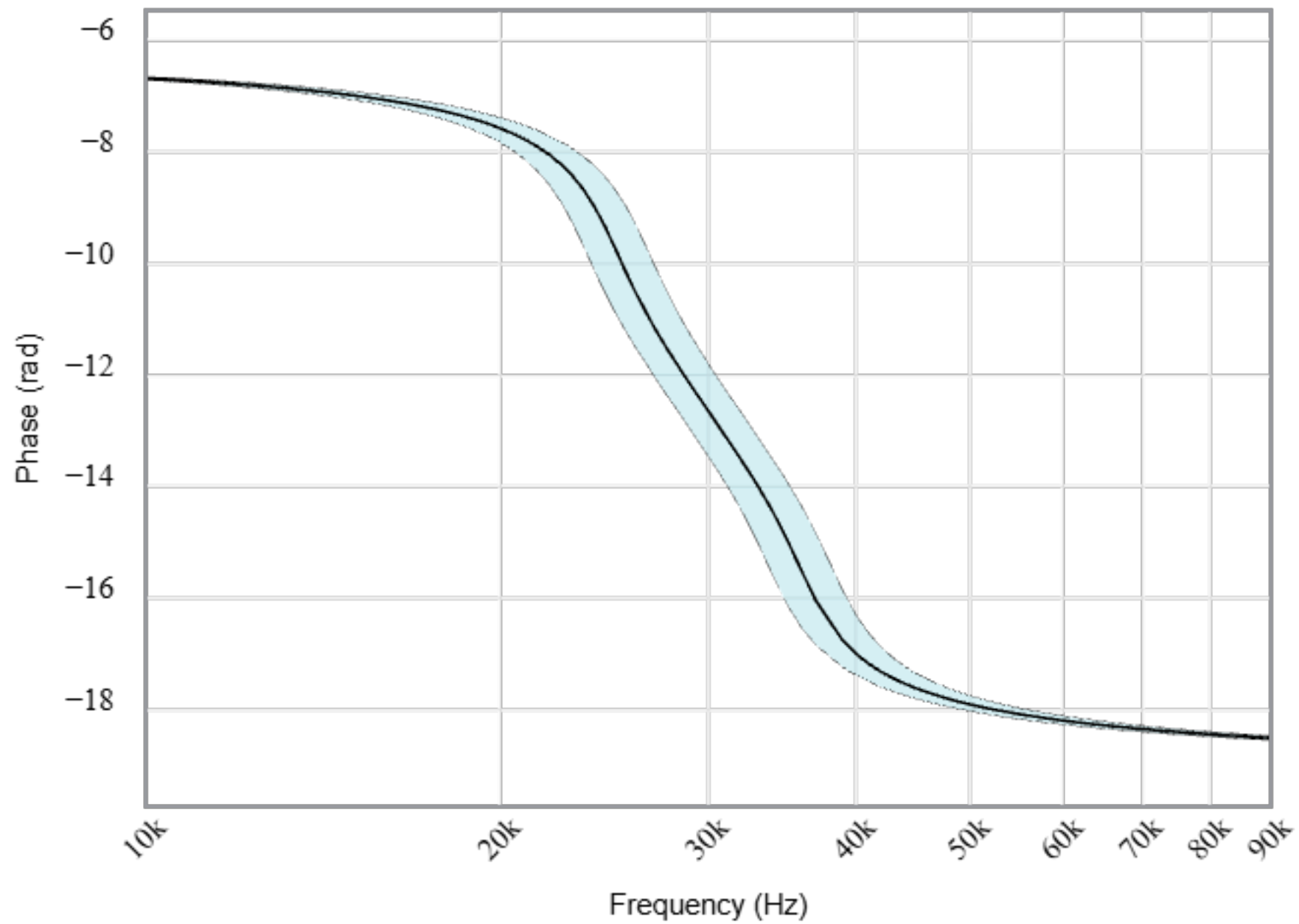
Magnitude(Volts per Volt)



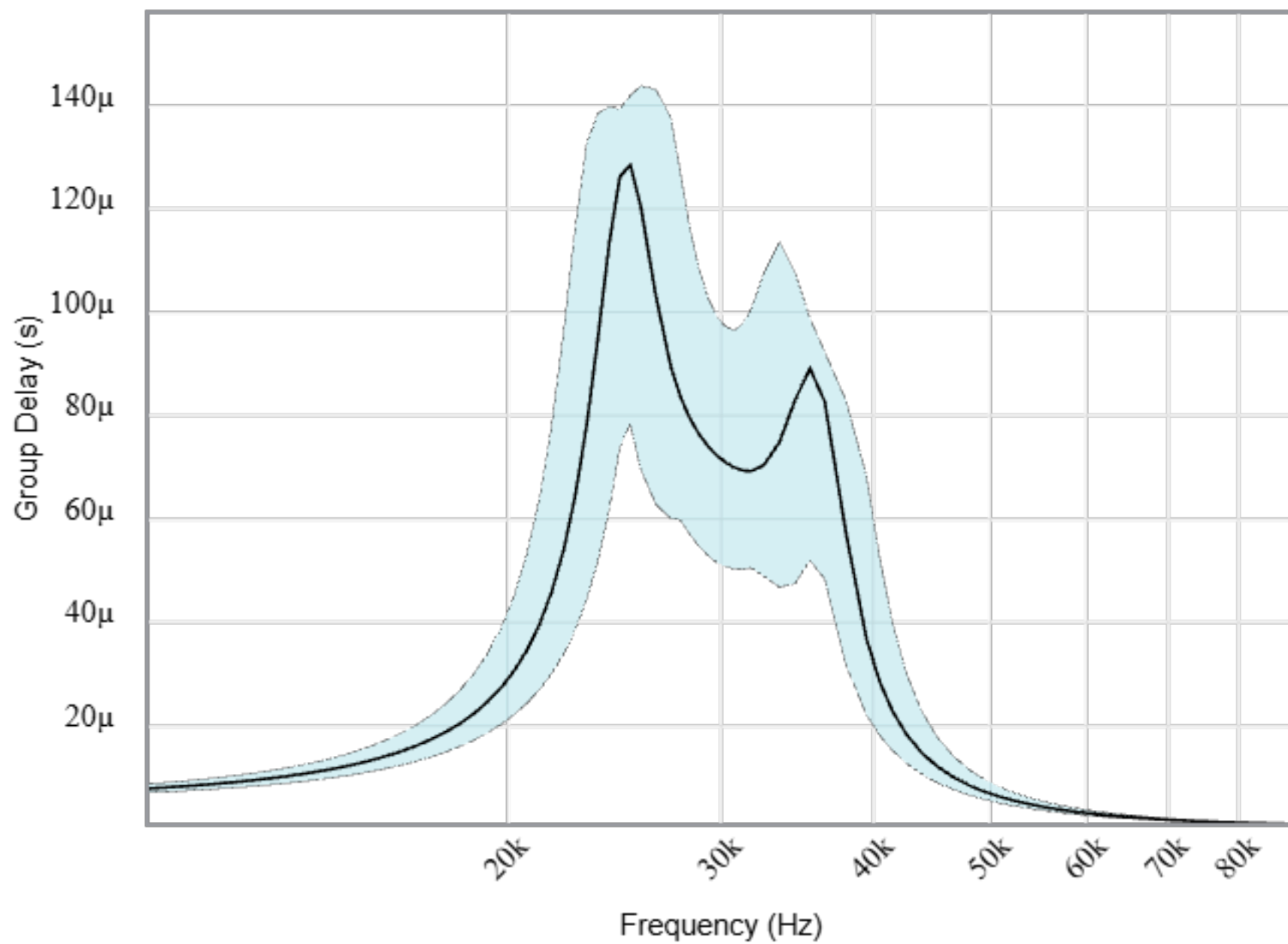
Phase(degrees)



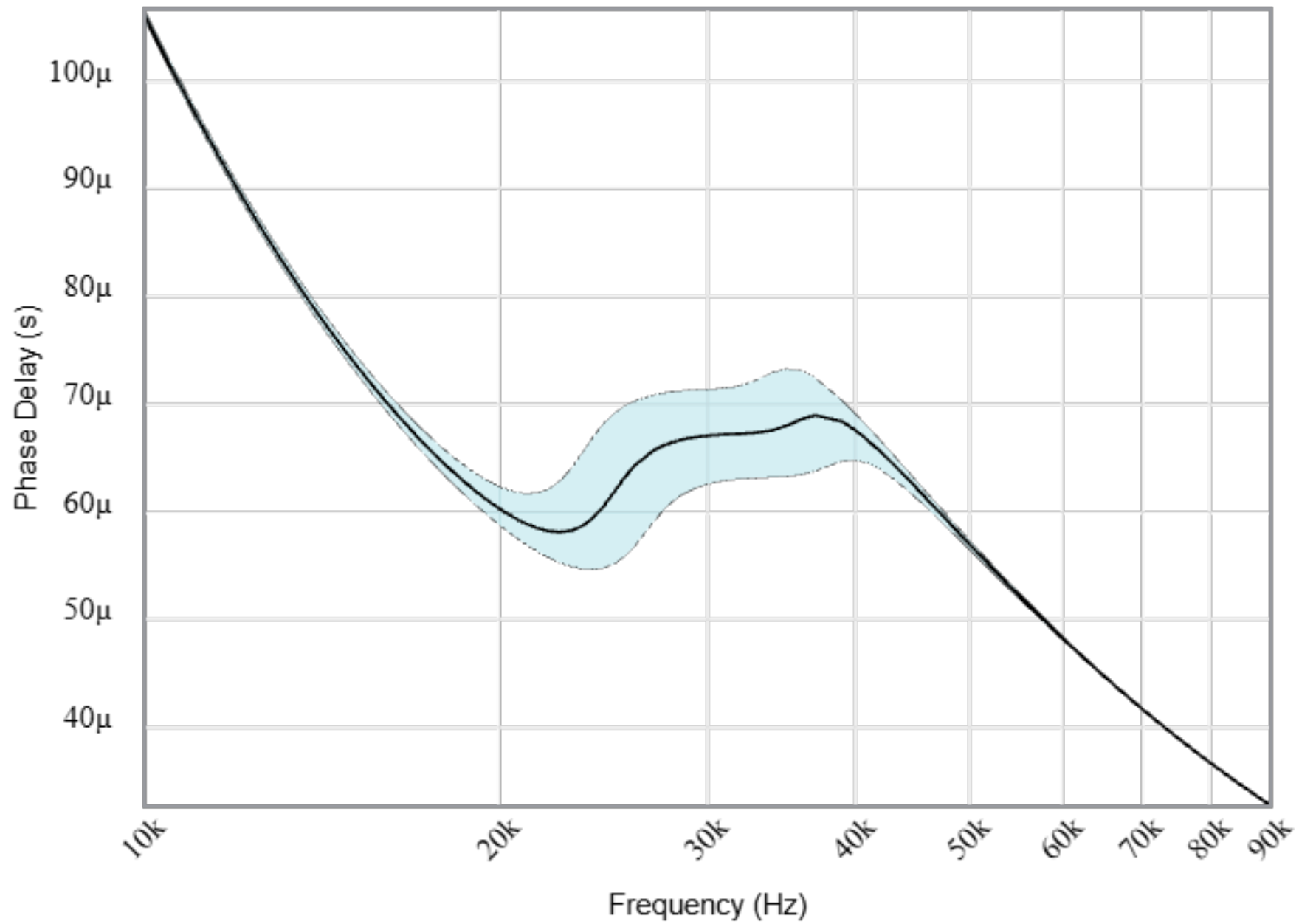
Phase(radians)



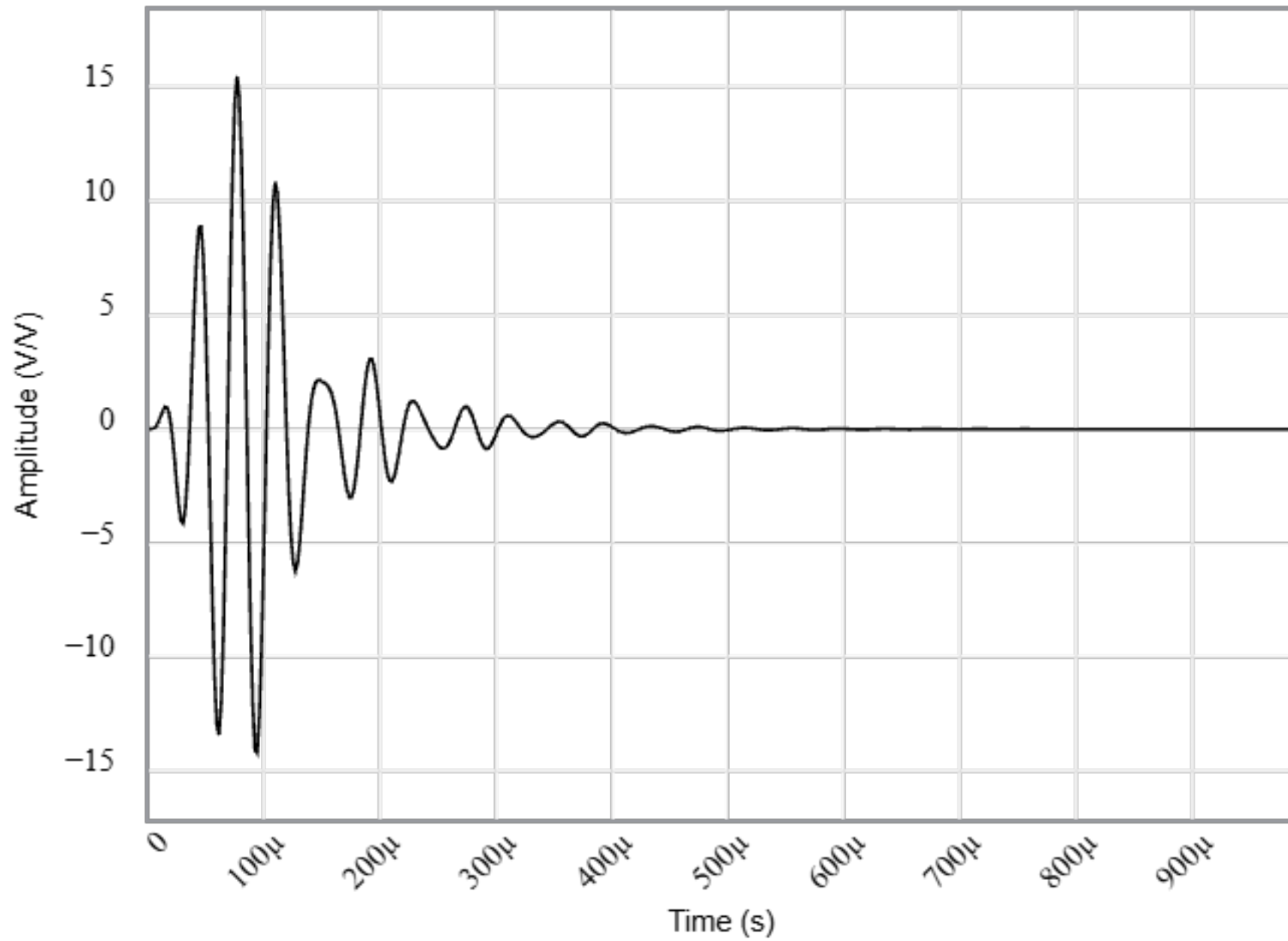
Group Delay



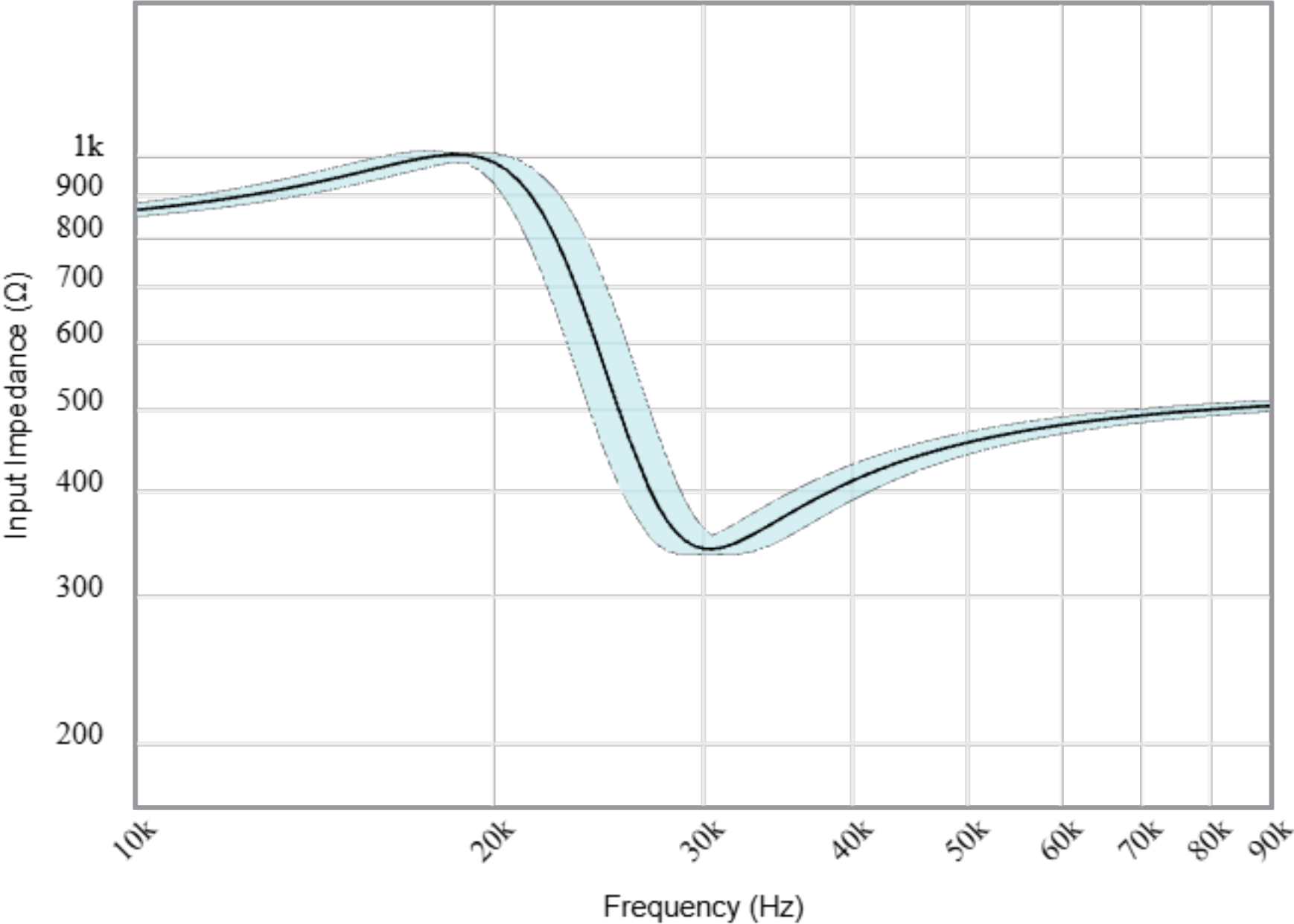
Phase Delay



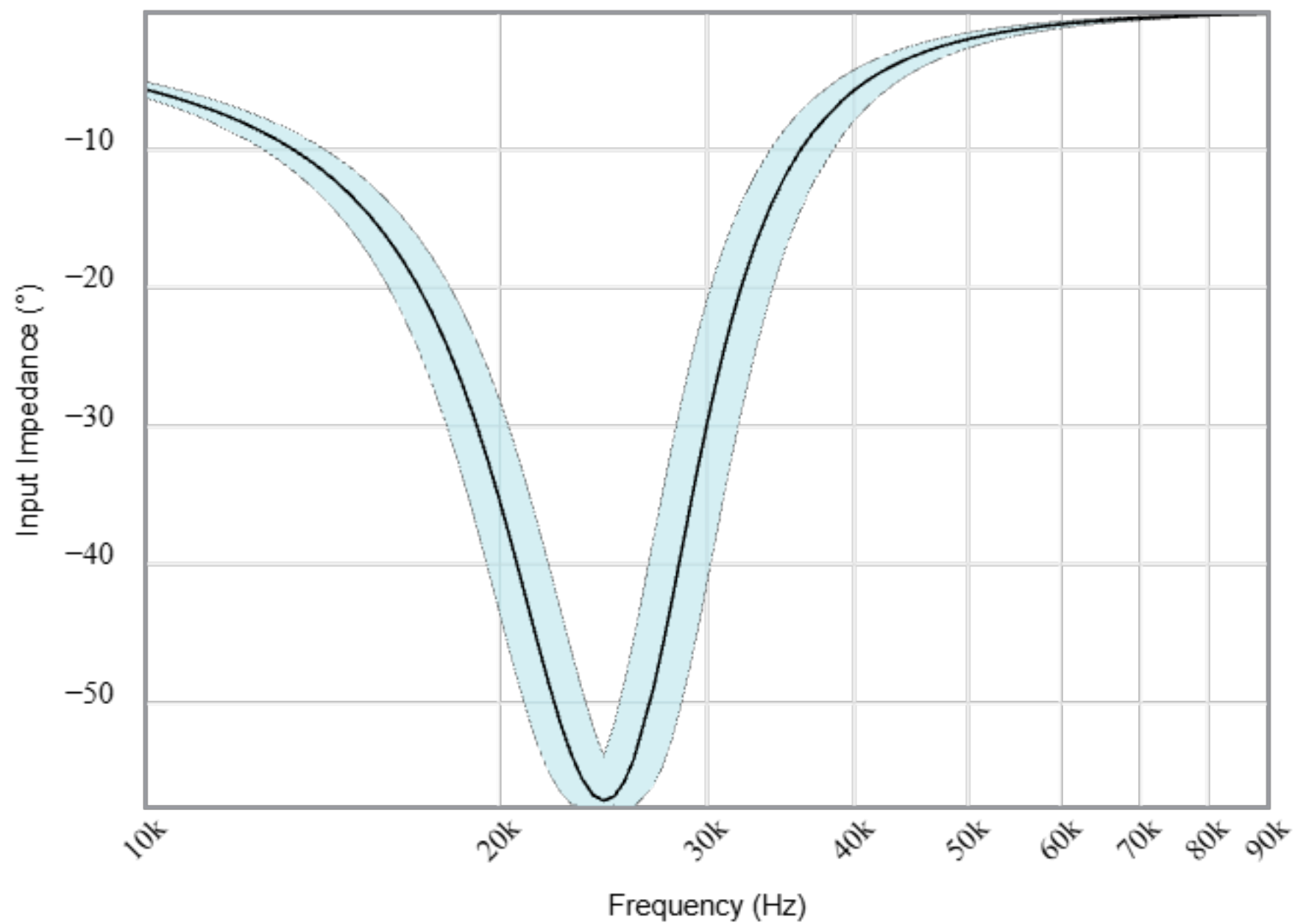
Step Response



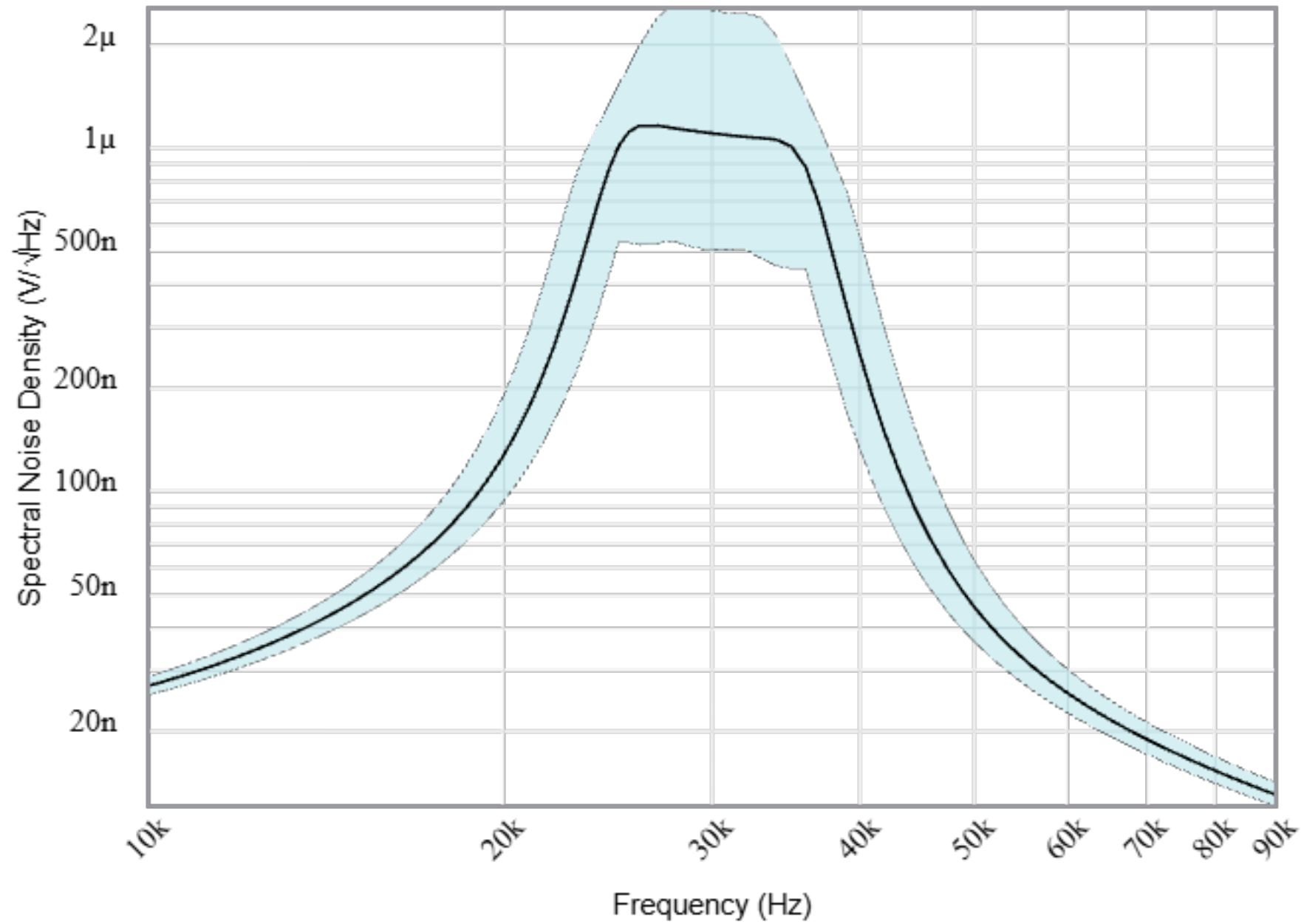
Input Impedance Magnitude



Input Impedance Phase



Noise



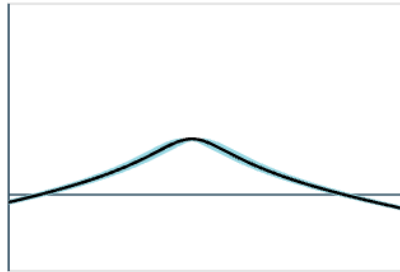
Stages

Your filter requires 4 op amp stage(s) with the following characteristics



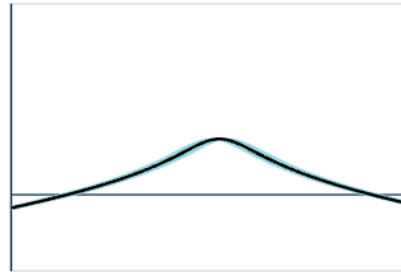
**2nd order
Band-Pass
Multiple
Feedback**

	Target	Simulated
Gain (V/V):	5.37	5.28 to 5.49
f_p (Hz):	27.8k	26.2k to 29.5k
Q:	2.75	2.74 to 2.8



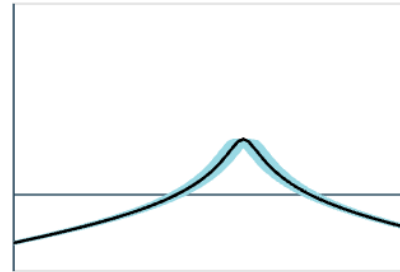
**2nd order
Band-Pass
Multiple
Feedback**

	Target	Simulated
Gain (V/V):	5.37	5.28 to 5.49
f_p (Hz):	32.4k	30.1k to 34k
Q:	2.75	2.74 to 2.8



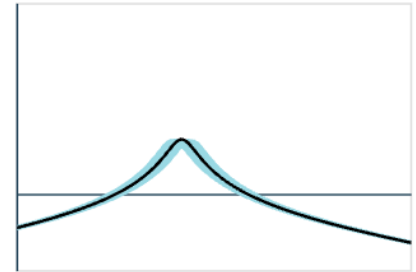
**2nd order
Band-Pass
Multiple
Feedback**

	Target	Simulated
Gain (V/V):	5.37	5.24 to 5.45
f_p (Hz):	36k	33.9k to 38.3k
Q:	6.73	6.81 to 6.93



**2nd order
Band-Pass
Multiple
Feedback**

	Target	Simulated
Gain (V/V):	5.37	5.22 to 5.44
f_p (Hz):	25k	23.6k to 26.6k
Q:	6.73	6.76 to 6.91



Circuit

